

Ruizhi Zhan

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Education



Ph.D. in Atmospheric Sciences

Department of Atmospheric and Oceanic Sciences, School of Physics, Peking University

September 2022 - June 2027, *expected*

Beijing, China

- Advisor: Prof. Daniel D.B. Koll



Visiting Ph.D. Student

Institute of Astronomy, University of Cambridge

May 2026 - November 2026

Cambridge, United Kingdom



B.S. in Physics

School of Physics and Technology, Wuhan University

September 2018 - June 2022

Wuhan, Hubei province, China

- Cumulative GPA: 3.87/4.00, rank 2/27

Publications

Ruizhi Zhan and Daniel D.B. Koll. Reinterpreting the JWST Observations of 55 Cancri e with a Non-grey General Circulation Model, *in press*, arXiv:2606.12116

Ruizhi Zhan and Daniel D.B. Koll. Are Observationally Allowed Thin Atmospheres Stable on Hot Rocky Exoplanets? *in prep.*

Ruizhi Zhan, Daniel D.B. Koll and Feng Ding. Novel Atmospheric Dynamics shape Inner Edge of Habitable Zone around White Dwarfs, *The Astrophysical Journal*, 971, 125, <https://iopscience.iop.org/article/10.3847/1538-4357/ad54c1>

Selected Awards

Cushman & Wakefield Scholarship

30 students per year

October 2024

Peking University

Outstanding Undergraduate Graduate

top 10%

June 2022

Wuhan University

National Scholarship

top 1%

November 2021

Wuhan University

First-class Scholarship of Wuhan University

top 5%

October 2021

Wuhan University

Talks & Presentations

Zhan R, D.D.B. Koll. Reinterpreting the JWST Observations of 55 Cancri e with a Non-Grey General Circulation Model.

Seminar talk

Exocoffee

MPIA, Heidelberg, Germany, 23 June 2026

Zhan R, D.D.B. Koll. Reinterpreting the JWST Observations of 55 Cancri e with a Non-Grey General Circulation Model.

Oral presentation

Rocky Worlds 4 conference

Groningen, Netherlands, 21 January 2026

Zhan R, D.D.B. Koll. Modeling the Atmosphere of 55 Cancri e with a Non-grey General Circulation Model.

Oral presentation

EGU 2025 conference

Vienna, Austria, 1 May 2025

Zhan R, D.D.B. Koll. Non-Grey Global Climate Model Reveals Atmospheric Heat Redistribution on Hot Rocky Exoplanets.

Oral presentation

AGU 2024 conference

Washington D.C., U.S., 10 Dec, 2024

Zhan R, D.D.B. Koll. Modeling the Atmosphere of 55 Cancri e with a Non-grey General Circulation Model.

Oral presentation

PhD Student Forum: Frontiers in Modern Astronomy

Beijing, China, 8 Dec, 2024

Zhan R, D.D.B. Koll and F. Ding. Novel Atmospheric Dynamics shape Inner Edge of Habitable Zone around White Dwarfs.

Oral presentation

2024 China Planetary Science Conference

Nanjing, China, 10 Oct, 2024

Zhan R, D.D.B. Koll and F. Ding. Novel Atmospheric Dynamics shape Inner Edge of Habitable Zone around White Dwarfs. *Exoplanets & Planet Formation Workshop 2023*
 Poster presentation *Beijing, China, 16 Dec, 2023*

Zhan R, D.D.B. Koll. Novel Atmospheric Dynamics shape Inner Edge of Habitable Zone around White Dwarfs. *Exoclines VI conference*
 Poster presentation *Exeter, U.K., 29 Jun 2023*

Zhan R, D.D.B. Koll. The Habitable Zone Inner Edge Around White Dwarfs. *AGU 2022 conference*
 Poster presentation *online, 16 Dec 2022*

Teaching

Teaching Assistant, Fundamental Planetary Science *Peking University*
Fall 2023

Teaching Assistant, Fluid Dynamics *Peking University*
Spring 2024

Competitive Observation Programs

(Co-I) *A Search for Life Around Two Dead Stars.*
 James Webb Space Telescope (JWST) Cycle 4 Guest Observers Program 7564
 Principal Investigator: Dr. Limbach, Mary Anne

Skills

Programming Python (Extensively) | LaTeX (Extensively) | shell | C++ | Fortran | Typst
Languages Mandarin Chinese (native) | English (Proficient)
Climate models ExoCAM | ExoRT | Exo-k | Isca | SOCRATES
Tools NASA-PSG

Service

Distinguished Lectures on Planetary Sciences “Atmospheric Dynamics of Giant Planets, Terrestrial Planets, and Earth” *Peking University*
 Local Organising Committee (LOC) Member *Aug 2023*

Atmospheres, Climates, and Habitability of Mars and Other Planets *Peking University*
 Local Organising Committee (LOC) Member *Aug 2024*

Atmosphere, Ocean, and Climate Dynamics of Earth and Other Planets *Peking University*
 Local Organising Committee (LOC) Member *Aug 2025*